

Lesson 1: Introduction to Bioprocesses

Learning objectives

- Define a bioprocess.
- Describe the three main stages of a bioprocess.

1.1 What is a bioprocess?

A bioprocess uses living cells (such as bacteria, yeast or fungi) or their enzymes to make useful products. Bread, yoghurt, vinegar, antibiotics and biofuels are all products of bioprocesses.

1.2 Stages of a bioprocess

Upstream processing prepares the raw materials (the medium), selects and grows the microorganism and sterilises the equipment.

Fermentation or bioconversion is the main reaction stage, carried out in a vessel called a bioreactor or fermenter under controlled conditions.

Downstream processing separates, purifies and packages the final product.

Stage	Purpose	Examples of steps
Upstream	Prepare medium and inoculum	Mixing, sterilising, culturing
Fermentation	Cells make the product	Controlling pH, temperature, oxygen
Downstream	Recover and purify product	Filtering, centrifuging, drying

Key points

- Bioprocesses use living cells or enzymes.
- The three stages are upstream, fermentation and downstream.
- Control of conditions is essential for good yield.

Review questions

- 1 Define the term bioprocess.
- 2 Give two products made by bioprocesses.
- 3 What happens during downstream processing?

Lesson 2: Microorganisms in Industry

Learning objectives

- Name useful industrial microorganisms.
- Match microorganisms with the products they make.

2.1 Choosing a microorganism

Industrial microorganisms should grow quickly on cheap raw materials, be safe to handle, give a high yield of the product and be easy to keep in pure culture.

2.2 Common examples

Yeast (*Saccharomyces cerevisiae*) ferments sugar into ethanol and carbon dioxide, used for bread, beer and wine.

Lactic acid bacteria convert milk sugar into lactic acid to make yoghurt and cheese.

Acetic acid bacteria turn ethanol into acetic acid, the main part of vinegar.

Moulds such as *Penicillium* produce antibiotics, and *Aspergillus* produces citric acid and enzymes.

Microorganism	Type	Product
<i>Saccharomyces cerevisiae</i>	Yeast	Bread, ethanol
<i>Lactobacillus</i>	Bacterium	Yoghurt, curd
<i>Acetobacter</i>	Bacterium	Vinegar
<i>Penicillium</i>	Mould	Penicillin
<i>Aspergillus niger</i>	Mould	Citric acid

Key points

- A good industrial strain grows fast and gives high yield.
- Yeast makes ethanol and CO₂ from sugar.
- Different microorganisms make different products.

Review questions

- 1 State two qualities of a good industrial microorganism.
- 2 Which microorganism is used in curd making?
- 3 What gas makes bread dough rise?

Lesson 3: Fermentation and Bioreactors

Learning objectives

- Describe the four phases of microbial growth.
- Compare batch, fed-batch and continuous fermentation.

3.1 The bioreactor

A bioreactor is a closed vessel where conditions such as temperature, pH, oxygen supply and stirring are monitored and controlled so that the microorganisms can grow and make the product efficiently.

3.2 Microbial growth curve

When microorganisms are placed in fresh medium they pass through four phases. During the **lag phase** cells adapt and do not multiply much. In the **log (exponential) phase** they multiply rapidly. In the **stationary phase** nutrients run low and growth stops balancing death. In the **death phase** more cells die than are produced.

3.3 Modes of operation

In **batch** fermentation everything is added at the start and the product is removed at the end. In **fed-batch** fermentation nutrients are added gradually during the run. In **continuous** fermentation fresh medium is fed in and product is removed all the time, so the process can run for long periods.

Mode	Feed	Product removal	Notes
Batch	All at start	At the end	Simple, easy to sterilise
Fed-batch	Added gradually	At the end	Avoids too much substrate
Continuous	Added continuously	Continuously	High output, harder to control

Key points

- Growth phases: lag, log, stationary, death.
- Most products are collected near the end of the log or in the stationary phase.
- Sterile conditions prevent contamination.

Review questions

- 1 List the four phases of a microbial growth curve in order.
- 2 State one difference between batch and continuous fermentation.
- 3 Why must a bioreactor be sterilised before use?